Internal Architecture of 8085 uP

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Internal Architecture of Intel's 8085 Microprocessor

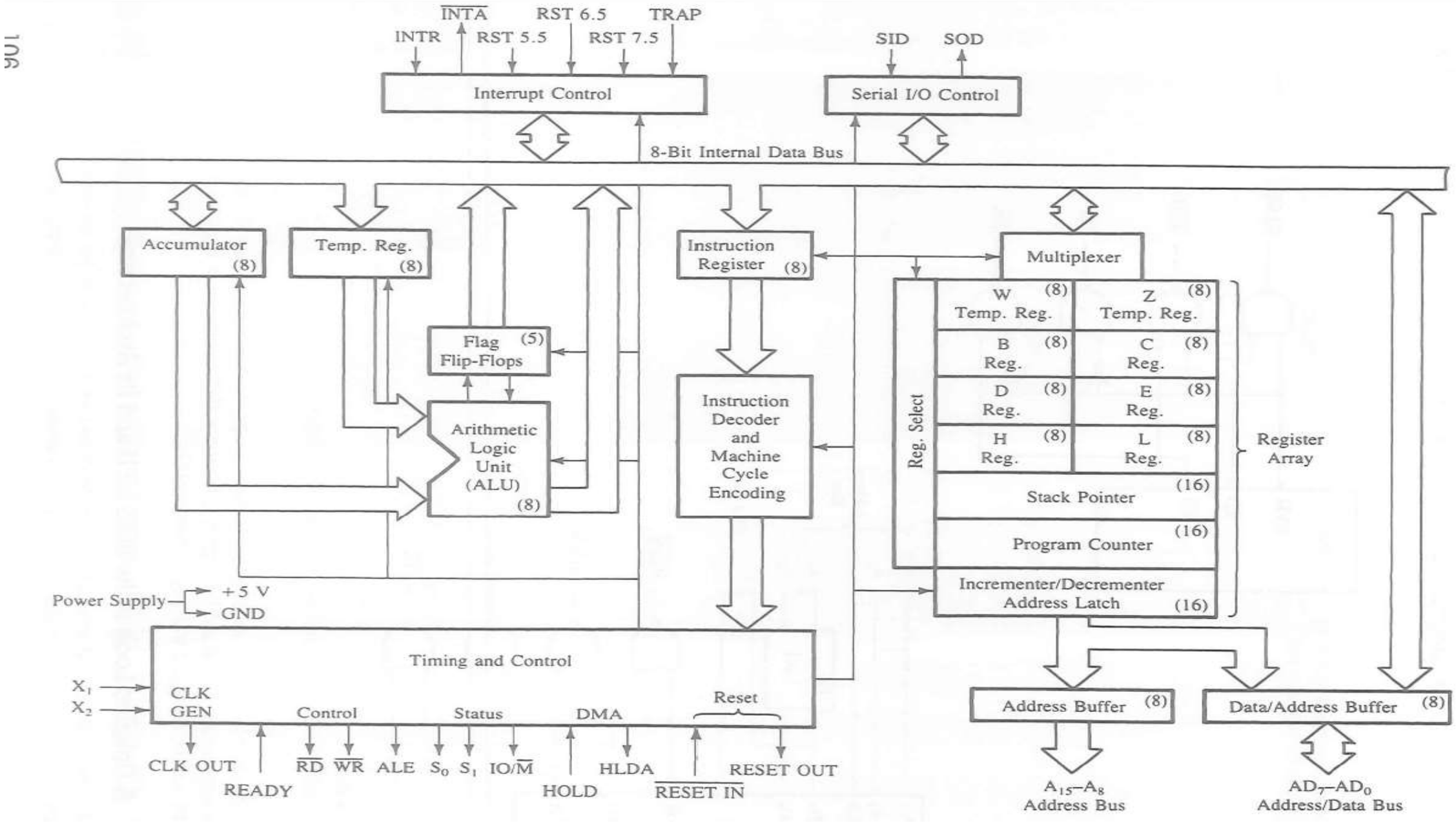
The Figure Shows the internal architecture of 8085 Microprocessor. It includes the ALU (Arithmetic/Logic Unit), Timing and Control Unit, Instruction Register and Decoder, Register Array, Interrupt Control, and Serial I/O Control.

THE ALU

The arithmetic/logic unit performs the computing functions; it includes the accumulator, the temporary register, the arithmetic and logic circuits, and five flags. The temporary register is used to hold data during an arithmetic/logic operation. The result is stored in the accumulator, and the flags (flip-flops) are set or reset according to the result of the operation.

The flags are affected by the arithmetic and logic operations in the ALU. In most of these operations, the result is stored in the accumulator. Therefore, the flags generally reflect data conditions in the accumulator—with some exceptions. The descriptions and conditions of the flags are as follows:

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- **S—Sign flag:** After the execution of an arithmetic or logic operation, if bit D_7 of the result (usually in the accumulator) is 1, the Sign flag is set. This flag is used with signed numbers. In a given byte, if D_7 is 1, the number will be viewed as a negative number; if it is 0, the number will be considered positive. In arithmetic operations with signed numbers, bit D_7 is reserved for indicating the sign, and the remaining seven bits are used to represent the magnitude of a number. However, this flag is irrelevant for the operations of unsigned numbers. Therefore, for unsigned numbers, even if bit D_7 of a result is 1 and the flag is set, it does not mean the result is negative. (See Appendix A2 for a discussion of signed numbers.)
- **Z—Zero flag:** The Zero flag is set if the ALU operation results in 0, and the flag is reset if the result is not 0. This flag is modified by the results in the accumulator as well as in the other registers.
- **AC—Auxiliary Carry flag:** In an arithmetic operation, when a carry is generated by digit D_3 and passed on to digit D_4, the AC flag is set. The flag is used only internally for BCD (binary-coded decimal) operations and is not available for the programmer to change the sequence of a program with a jump instruction.
- **P—Parity flag:** After an arithmetic or logical operation, if the result has an even number of 1s, the flag is set. If it has an odd number of 1s, the flag is reset. (For example, the data byte 0000 0011 has even parity even if the magnitude of the number is odd.)
- **CY—Carry flag:** If an arithmetic operation results in a carry, the Carry flag is set; otherwise it is reset. The Carry flag also serves as a borrow flag for subtraction.

The bit positions reserved for these flags in the flag register are as follows:

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
S	Z		AC		P		CY

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TIMING AND CONTROL UNIT

This unit synchronizes all the microprocessor operations with the clock and generates the control signals necessary for communication between the microprocessor and peripherals. The control signals are similar to a sync pulse in an oscilloscope. The $\overline{RD}$ and $\overline{WR}$ signals are sync pulses indicating the availability of data on the data bus.

INSTRUCTION REGISTER AND DECODER

The instruction register and the decoder are part of the ALU. When an instruction is fetched from memory, it is loaded in the instruction register. The decoder decodes the instruction and establishes the sequence of events to follow. The instruction register is not programmable and cannot be accessed through any instruction.

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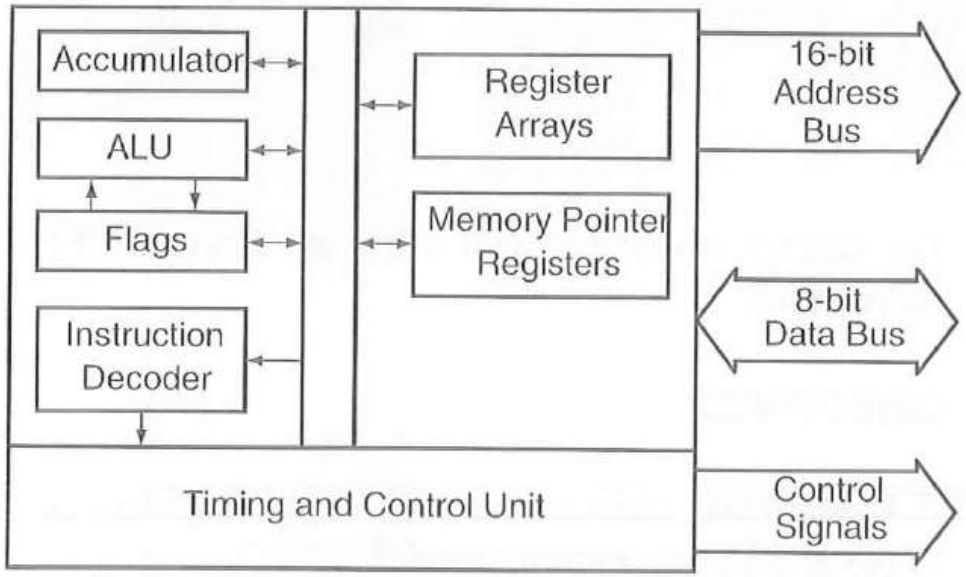
Registers

- **Accumulator** or A register is an 8-bit register used for arithmetic, logic, I/O and load/store operations.

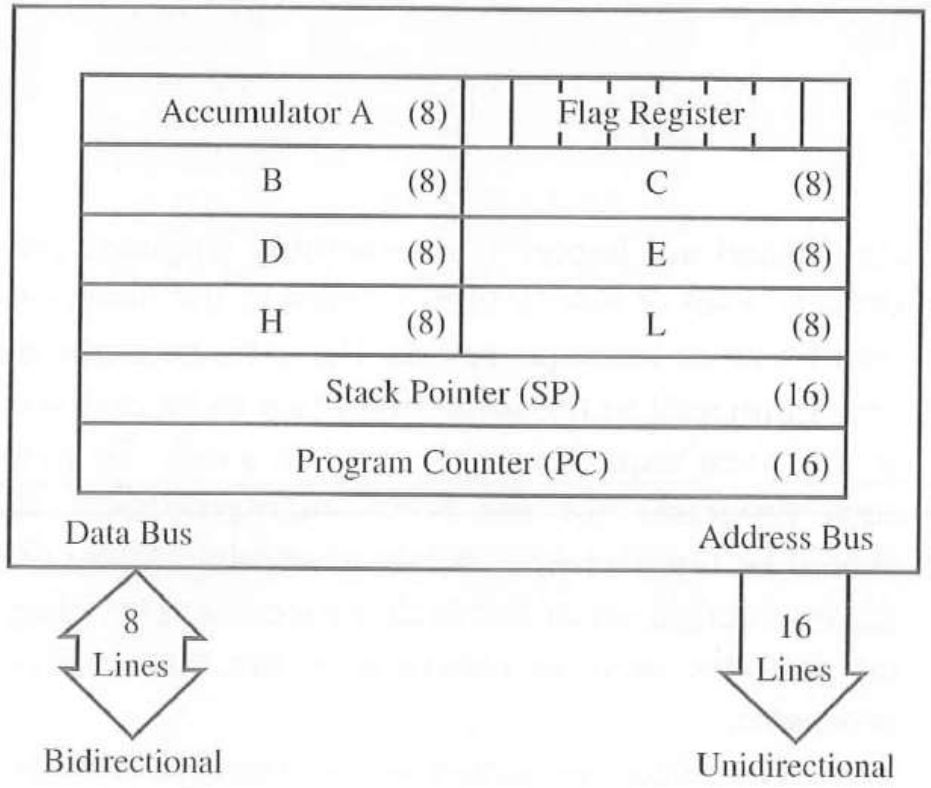
General Registers

- 8-bit B and 8-bit C registers can be used as one 16-bit BC register pair. When used as a pair the C register contains low-order byte. Some instructions may use BC register as a data pointer.
- 8-bit D and 8-bit E registers can be used as one 16-bit DE register pair. When used as a pair the E register contains low-order byte. Some instructions may use DE register as a data pointer.
- 8-bit H and 8-bit L registers can be used as one 16-bit HL register pair. When used as a pair the L register contains low-order byte. HL register usually contains a data pointer used to reference memory addresses.
- **Stack pointer** is a 16 bit register. This register is always decremented/incremented by 2 during push and pop.
- **Program counter** is a 16-bit register.

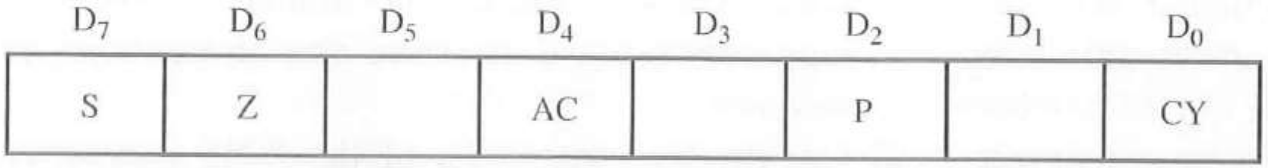
a)Hardware Model b)Programming Model & c) Flag Register of Intel's 8085 Microprocessor



(a)



(b)



(c)

a)Hardware Model b)Programming Model & c) Flag Register of Intel's 8085 Microprocessor

8085 Hardware Model

The hardware model in Figure (a) shows two major segments. One segment includes the arithmetic/logic unit (ALU) and an 8-bit register called an accumulator, instruction decoder, and flags. The second segment shows 8-bit and 16-bit registers. Both segments are connected with various internal connections called an internal bus. The arithmetic and logical operations are performed in the ALU. Results are stored in the accumulator, and flip-flops, called flags, are set or reset to reflect the results (see Figure a). There are three buses: a 16-bit unidirectional address bus, an 8-bit bidirectional data bus, and a control bus. these three buses were shown as one system bus. The 8085 processor uses the 16-bit address bus to send out memory addresses, the 8-bit data bus to transfer data, and the control bus for timing signals.

a)Hardware Model b)Programming Model & c) Flag Register of Intel's 8085 Microprocessor

8085 Programming Model

The programming model consists of some segments of the ALU and the registers. This model does not reflect the physical structure of the 8085 but includes the information that is critical in writing assembly language programs. The model includes six registers, one accumulator, and one flag register, as shown in Figure (b). In addition, it has two 16-bit registers: the stack pointer and the program counter. They are described briefly as follows.

REGISTERS

The 8085 has six general-purpose registers to store 8-bit data; these are identified as B, C, D, E, H, and L, as shown in Figure (b). They can be combined as register pairs—BC, DE, and HL—to perform some 16-bit operations. The programmer can use these registers to store or copy data into the registers by using data copy instructions.

ACCUMULATOR

The accumulator is an 8-bit register that is part of the arithmetic/logic unit (ALU). This register is used to store 8-bit data and to perform arithmetic and logical operations. The result of an operation is stored in the accumulator. The accumulator is also identified as register A.